



Graphle

Graph-oriented file management system

Organize your files the way you think.

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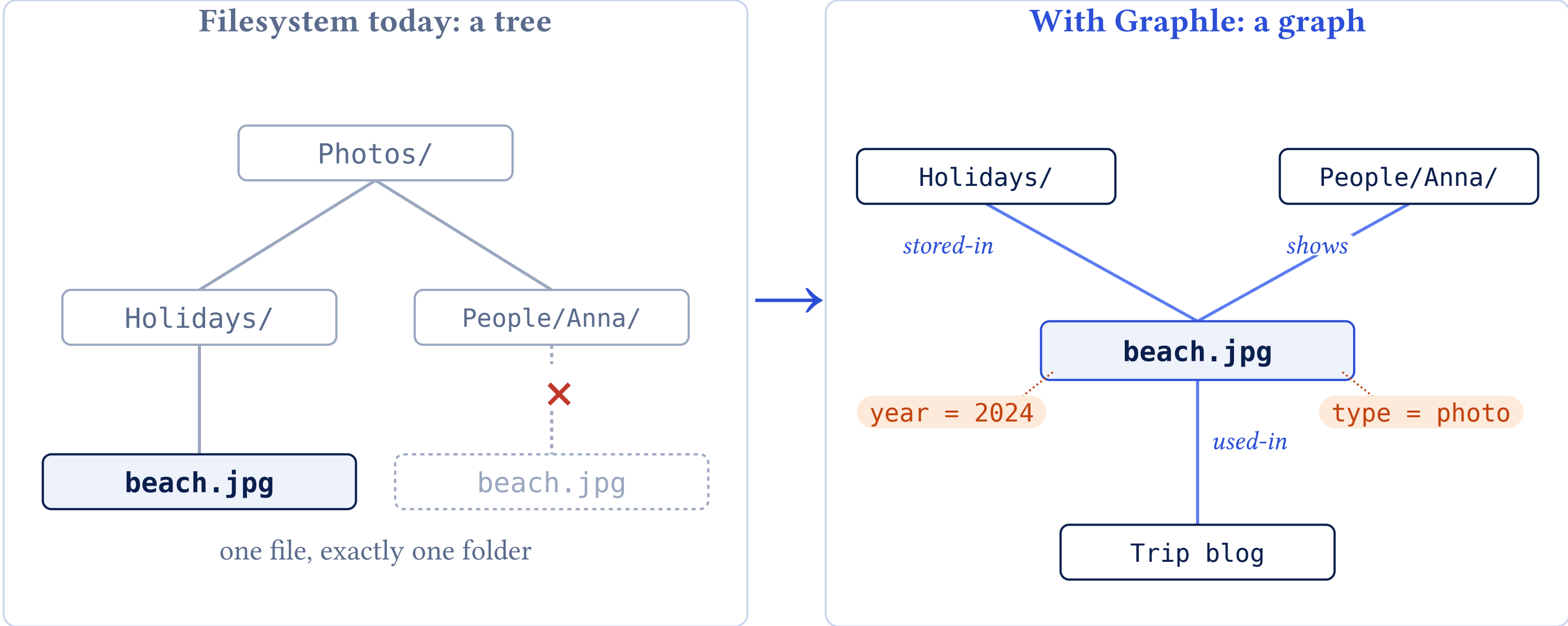
Motivation

A filesystem is a tree: every file lives in exactly one folder. Yet a single photo belongs to a holiday, a person and a project all at the same time. Graphle leaves files where they are and adds the connections a tree cannot express.

Goals

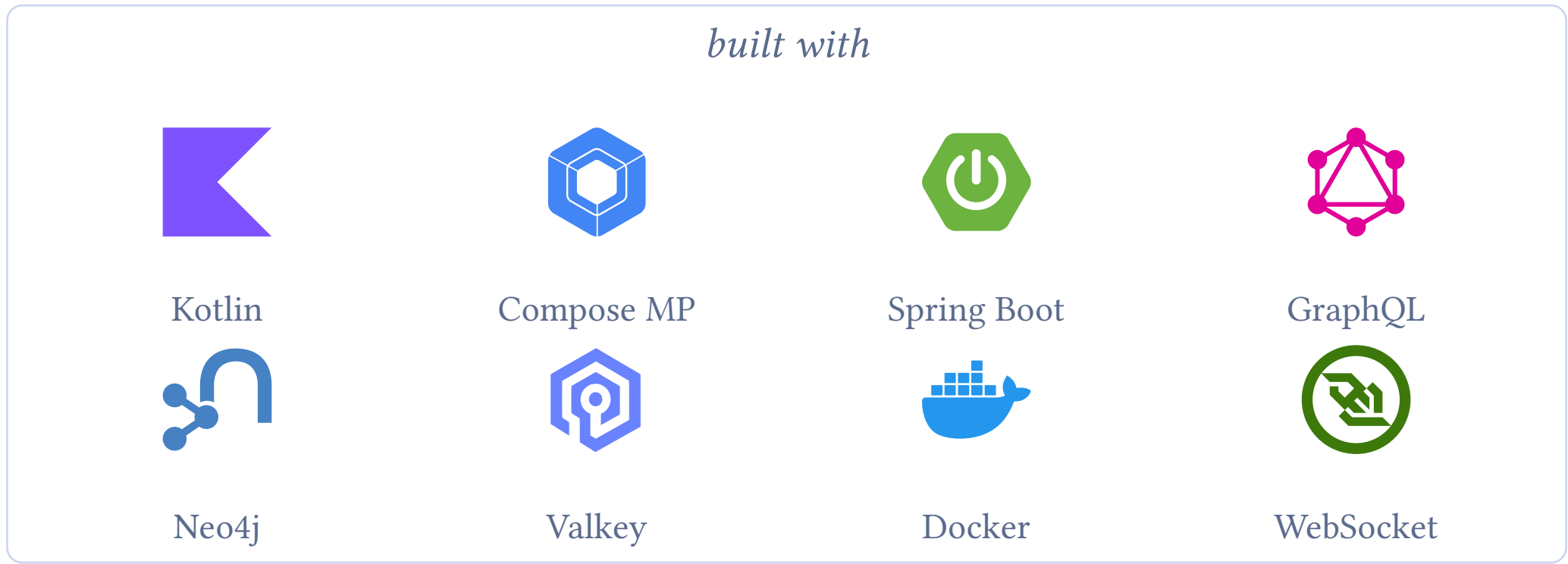
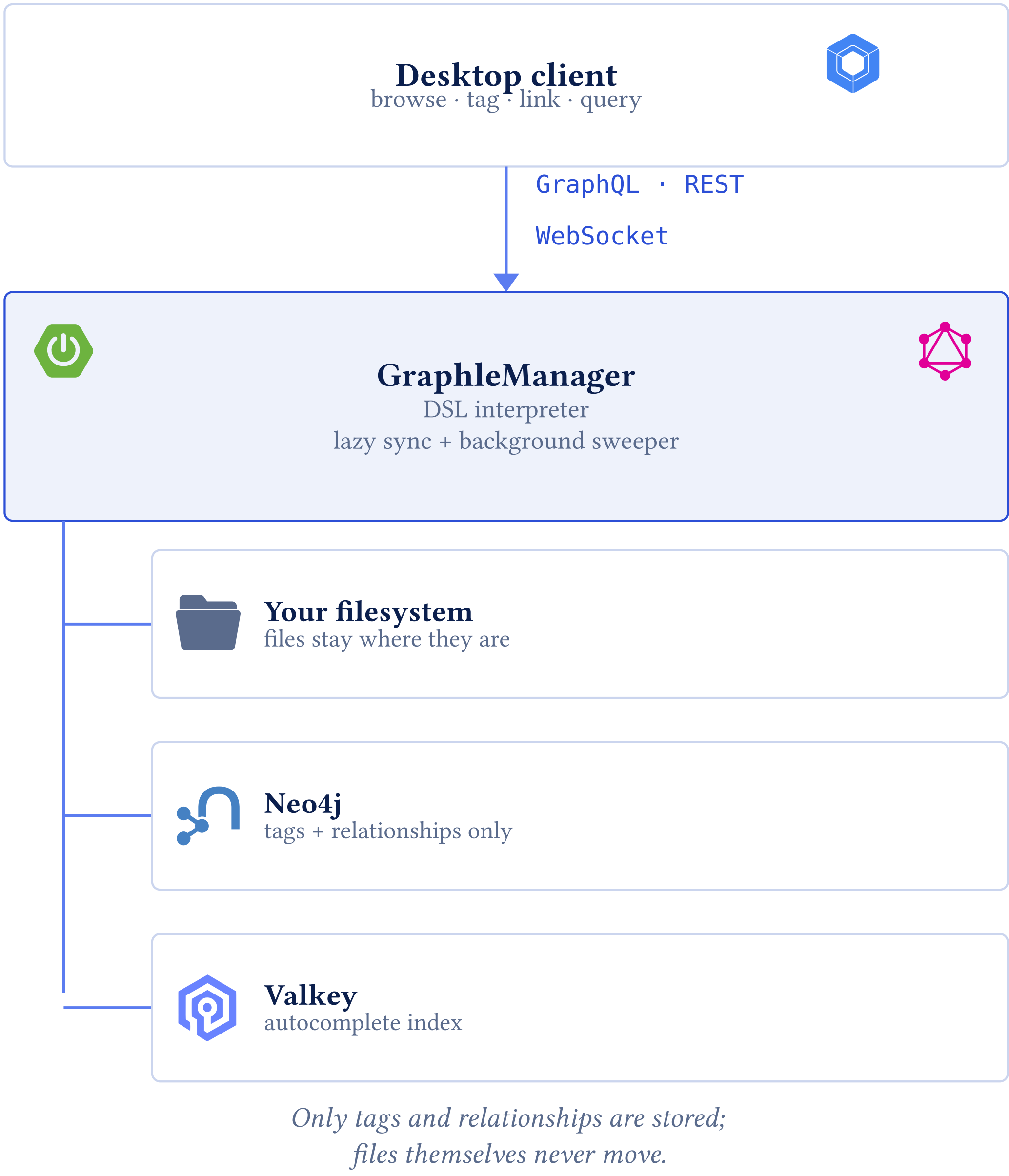
- **Graph data model** over the existing filesystem
- **Query language** for structure, tags and relationships
- **Backend** keeping the graph in sync with a live filesystem
- **Remote access** to a backend on another machine
- **Graphical client** for browsing, editing and querying

Why a graph?

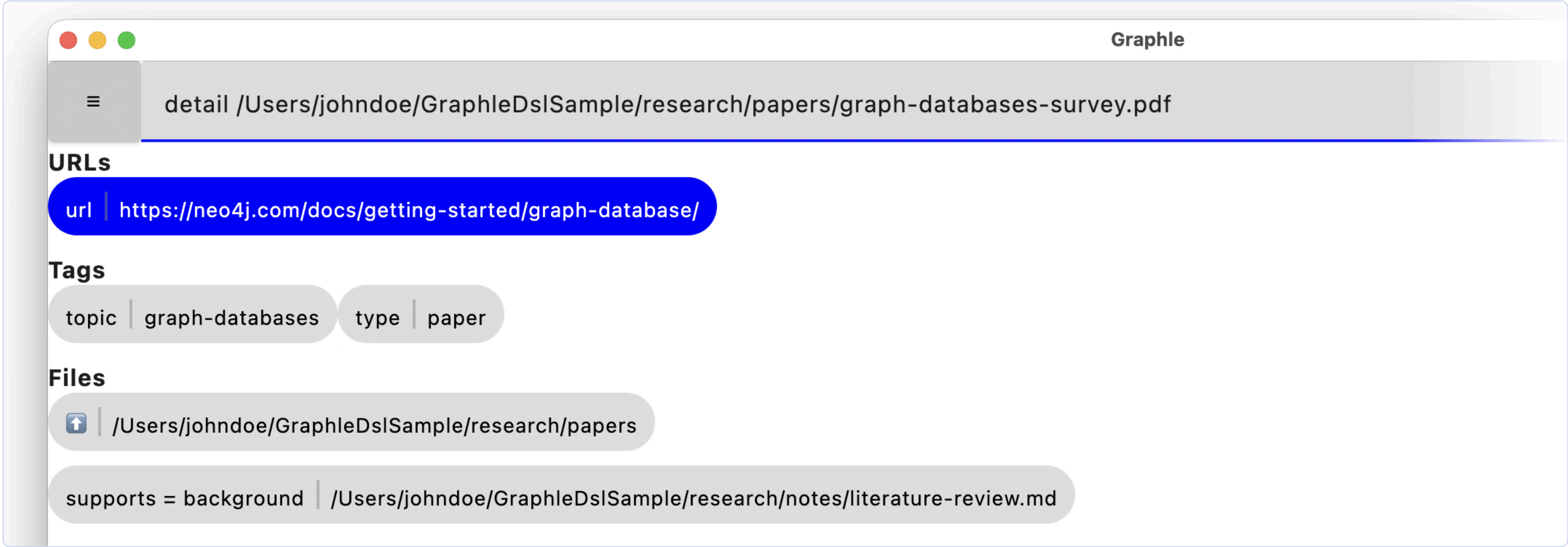


One photo, three contexts, plus tags no folder name could hold.

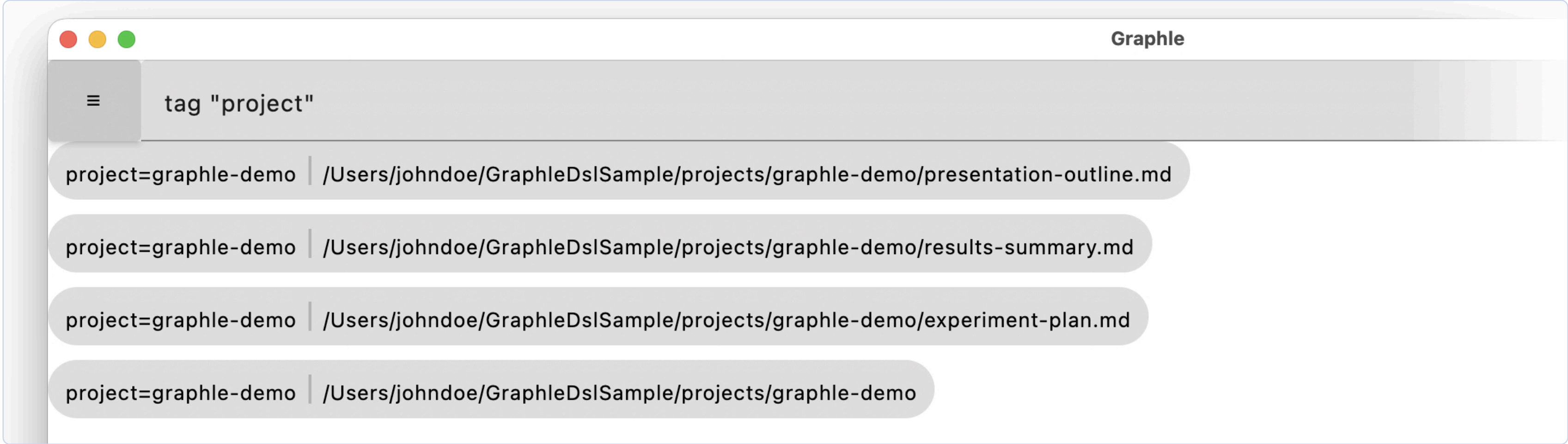
How it works



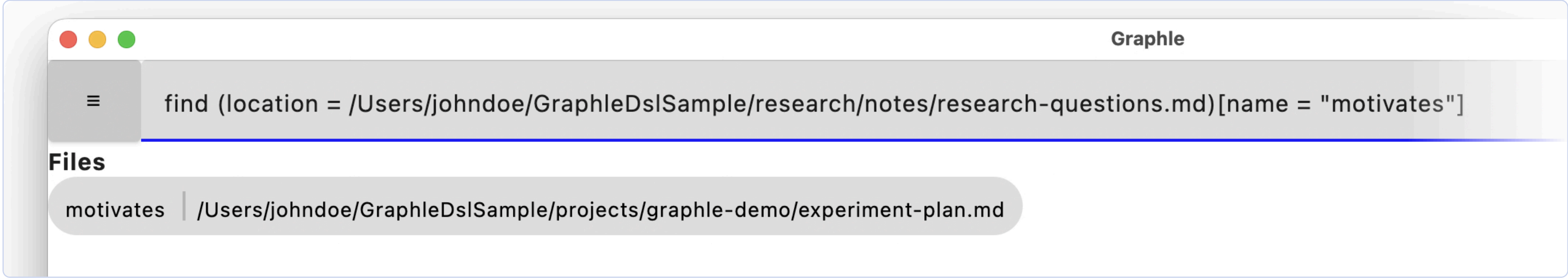
1 Open a file: its tags, links and relationships



2 Ask for a tag: every file that carries it



3 Follow a relationship: jump to what it points at



The command line always shows the query behind the view.

Query language

```
find (tagName = "topic") [name = "cites"] ()

      |
      | files carrying that tag
      |
      | follow this relationship
      |
      | to any file

tag a file addTag "the-matrix.mkv" "year" "1999"
link two files addRel "questions.md" "experiment.md" "motivates"
```

Acknowledgements

Thanks to my supervisor, Pavel Koupil, for his guidance throughout this thesis.

Conclusion

Files can stay in the filesystem and still be organised and explored as a graph. The result is a working cross-platform application: a graph data model, a query DSL, a backend exposing GraphQL, REST and WebSocket interfaces, and a desktop client.

Code & contact

GitHub

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